

Windows Build Status

What is proven to work on native Windows right now, what still needs a real machine to confirm, and the exact steps to build and run it.

01 Status

BUILD

Verified

`forward_search`'s C++/OpenCL code compiles and links on native Windows via MinGW-w64 (MSYS2), confirmed by GitHub Actions.

RUN

Not yet verified

GitHub's Windows runners do not reliably have a usable OpenCL device, so CI only proves the build compiles and links, not that it runs correctly against real data.

- ❗ The convenient single command workflow (`projection-center pipeline --backend cpp, run_all_128.sh`, and so on) has not been tested on Windows. Only `forward_search`'s own C++ build was attempted this session. `projection-center` is a separate Python package (with a `pyopencl` dependency) that has not been touched for Windows at all.

02 A real blocker found and fixed

`--backend cpp` used to always fail with "binary not found" on Windows, even after a successful build: `CppBackend` looked for the Linux binary names, no `.exe`, which never exist on Windows. Fixed to look for `forward_search.exe` / `forward_search_resample.exe` there instead.

Without `projection-center` confirmed working end to end, running the two `.exe` files directly still hits the pose format mismatch already documented for Linux: `search` writes an HDF5 pose file, `resample` expects JSON, they cannot be chained by hand, the JSON conversion only happens inside `projection-center`'s own CLI.

03 Building

Using MSYS2's MinGW64 environment, the same toolchain the CI workflow uses.

```
# from an MSYS2 MINGW64 shell
pacman -S --needed mingw-w64-x86_64-toolchain mingw-w64-x86_64-meson \
  mingw-w64-x86_64-ninja mingw-w64-x86_64-hdf5 mingw-w64-x86_64-boost \
  mingw-w64-x86_64-opengl-headers mingw-w64-x86_64-opengl-icd

# MSYS2's opengl-headers package does not include the C++ bindings header
# forward_search/ uses, fetch it once, matching the CI workflow:
curl -sSL -o /mingw64/include/CL/opengl.hpp \
  https://raw.githubusercontent.com/KhronosGroup/OpenCL-CLHPP/main/include/CL/opengl.hpp

cd forward_search
meson setup builddir
meson compile -C builddir
```

Produces `builddir/forward_search.exe` and `builddir/forward_search_resample.exe`, same as `forward_search` and `forward_search_resample` on Linux.

04 Running (search only)

Same flags as Linux, per `RUNNING.md`.

```
./builddir/forward_search.exe --data path\to\data.hdf5 --mode buffer \
  --kernel kernels/forward_search.cl --output pose.h5
```

Prints the same MSE: / xshift: / alpha: / beta: summary as the Linux binary, assuming a real OpenCL device is present.

05 Full pipeline (search + resample, one command)

Needs `projection-center` installed (`pip install -e projection-center`).

```
projection-center pipeline --backend cpp --cpp-mode buffer --data data.hdf5 \
  --output-pose pose.json --output-data resampled.hdf5
```

Not yet confirmed working on Windows. CI now attempts exactly this command against a small synthetic dataset, that is the next thing to check.

06 What is still open

- `run_all_128.sh` / `run_all_512.sh` call `python3`, which a plain Windows Python install usually does not provide, just `python`. Would need fixing, and need Git Bash or MSYS2 to run at all.
- Built with MinGW-w64, not MSVC. Switching to MSVC would need further flag changes (different optimization and linker syntax) not attempted yet.